IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

WAVELETS, REPRESENTATION LEARNING AND THEIR APPLICATIONS

Thursday, 14 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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EXAMINATIONS 2026

WAVELETS AND APPLICATIONS

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Special Information for the Invigilators: NONE

Information for Candidates:

Sub-sampling by an integer N

$$x_{\downarrow N}[n] \Longleftrightarrow \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-2\pi k)/N}) = \frac{1}{N}\sum_{k=0}^{N-1} X(W_N^k z^{1/N}),$$

where

$$W_N^k = e^{-j2\pi k/N}.$$

Dual Basis:
Given a basis $\{\varphi_i(t)\}_{i\in\mathbb{Z}}$, the dual basis is given by the set of elements $\{\tilde{\varphi}_i(t)\}_{i\in\mathbb{Z}}$ satisfying:

$$\langle \varphi_i(t), \tilde{\varphi}_j(t)\rangle = \delta_{i,j}.$$

Haar scaling and wavelet functions
The Haar scaling function is defined as:

$$\varphi(t) = \begin{cases} 1 & 0 \le t < 1 \\ \\ 0 & \text{otherwise} \end{cases}$$

The Haar wavelet is defined as:

$$\psi(t) = \begin{cases} 1 & 0 \le t < 1/2 \\ \\ -1 & 1/2 \le t < 1 \\ \\ 0 & \text{otherwise} \end{cases}$$

A function $f(t) \in L_2(\mathbb{R})$ if

$$\int_{-\infty}^{\infty} |f(t)|^2 dt < \infty.$$

A sequence $f[n] \in \ell_2(\mathbb{Z})$ if

$$\sum_{n=\infty}^{\infty} |f[n]|^2 < \infty.$$

The Fourier transform of the B-spline of order N is

$$\hat{\beta}_N(\omega) = e^{-jN\omega/2}\left(\frac{\sin\omega/2}{\omega/2}\right)^N.$$

The Questions

1. Consider the two-channel filter bank of Figure 1a with analysis filters $H_0(z)$ and $H_1(z)$, and synthesis filters $G_0(z)$ and $G_1(z)$. We seek to design an orthogonal FIR filter bank

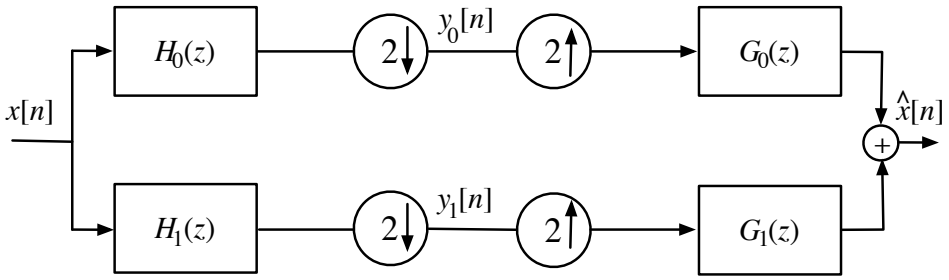


Figure 1a: Two-channel filter bank.

where the filters have length $L = 4$.

(a) Let the lowpass synthesis filter be $G_0(z) = g_0 + g_1 z^{-1} + g_2 z^{-2} + g_3 z^{-3}$. Given the orthonormality condition $\sum_n g_n g_{n-2k} = \delta_k$ and assuming that you want the filter to have two zeros at $\omega = \pi$, show that the coefficients for the resulting 4-tap filter are:

$$g_0 = \frac{1+\sqrt{3}}{4\sqrt{2}}, \quad g_1 = \frac{3+\sqrt{3}}{4\sqrt{2}}, \quad g_2 = \frac{3-\sqrt{3}}{4\sqrt{2}}, \quad g_3 = \frac{1-\sqrt{3}}{4\sqrt{2}}.$$

[10]

(b) Using the coefficients from part (a), derive the exact expression for the filters $H_0(z)$, $H_1(z)$ and $G_1(z)$ that would lead to a perfect-reconstruction orthogonal filter bank. [6]

(c) Assume that $x[n] = 1 + n^2$ and ignore any boundary effect. Which of the signals $y_0[n]$ and $y_1[n]$ is nonzero? (Justify your answer). [3]

(d) Assume now that $x[n] = n + 2$ and ignore again any boundary effect. Which of the signals $y_0[n]$ and $y_1[n]$ is nonzero? (Justify your answer). [3]

(e) Suppose you were given a two-channel FIR filter bank with real coefficients and lowpass synthesis filter

$$G_0(z) = g_0 + g_1 z^{-1} + g_2 z^{-2} + g_3 z^{-3} + g_4 z^{-4}.$$

Could this be an orthogonal filter bank with $G_0(z)$ having three zeros at $\omega = \pi$? Briefly justify your answer. [3]

2. Let $\varphi(t)$ be the Haar scaling function and $\psi(t)$ be the Haar wavelet function.

(a) Given a piecewise constant signal $f(t) = \sum_{n=-\infty}^{\infty} f[n]\varphi(t-n)$. Show that $f(t)$ is in $L_2(\mathbb{R})$ if and only if $f[n]$ is in $\ell_2(\mathbb{Z})$. [5]

(b) Now assume that

$$f(t) = 1 - \cos(2\pi t) \qquad t \in [0, 1],$$

i. compute the coefficients c_i of the orthogonal projection of $f(t)$ onto the space spanned by $\varphi(t), \psi(t), \sqrt{2}\psi(2t)$ and $\sqrt{2}\psi(2t-1)$. [5]

ii. Sketch the orthogonal projection $\hat{f}(t)$. [5]

iii. Verify that the error function $\epsilon(t) = f(t) - \hat{f}(t)$ is orthogonal to the space spanned by $\varphi(t), \psi(t), \sqrt{2}\psi(2t)$ and $\sqrt{2}\psi(2t-1)$. [5]

(c) Assume now that $\varphi(t)$ is a B-spline of order N, that is $\varphi(t) = \beta_N(t)$. Show that

$$A(\omega) = \sum_k |\hat{\varphi}(\omega + 2\pi k)|^2 = \left(2 \sin \frac{\omega}{2}\right)^{2N} \sum_{k=-\infty}^{\infty} (\omega + 2\pi k)^{-2N}.$$

[Hint: use the fact that $\sin(\omega/2 + \pi k) = (-1)^k \sin(\omega/2)$.] [5]

3. Consider the two-channel filter bank of Figure 3a.

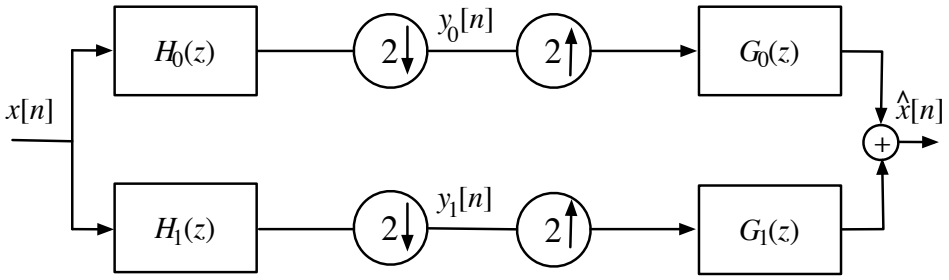


Figure 3a: Two-channel filter bank.

(a) Express $\hat{X}(z)$ as a function of $X(z)$ and the filters. Then derive the two perfect reconstruction (PR) conditions the filters have to satisfy. [5]

(b) Assume

$$P(z) = G_0(z)H_0(z) = \frac{1}{16}\left(-1 + 9z^{-2} + 16z^{-3} + 9z^{-4} - z^{-6}\right).$$

i. Find all the filters for this perfect-reconstruction filter bank if

$$H_0(z) = \left(\frac{1+z^{-1}}{2}\right)^3.$$

[10]

(c) Consider now the system

$$y = HD_2GHD_4HD_2Fx$$

where G, H and F are the infinite matrix representations for filtering and D_k is the matrix representation of downsampling by k with $k = 2, 4$.

i. Draw a block diagram of the system. [5]

ii. Derive an equivalent system consisting of a single filter block and a single downsampling matrix. Write the z-transform of the equivalent system as a function of $G(z)$, $H(z)$ and $F(z)$. [5]

4. The lifting scheme is a technique that can be used to implement a two-channel filter-bank decomposition. A lifting transform is built in stages. First, the signal $x[n]$ is split into even and odd parts. We denote with $E(z)$ the z-transform of the even component $x_e[n] = x[2n]$ and with $O(z)$ the z-transform of the odd component $x_o[n] = x[2n+1]$. The first stage then creates a detail signal $d[n]$ by prediction, and the second stage corrects the low-frequency part by an update to get $s[n]$. A simple schematic is shown in Figure 4a.

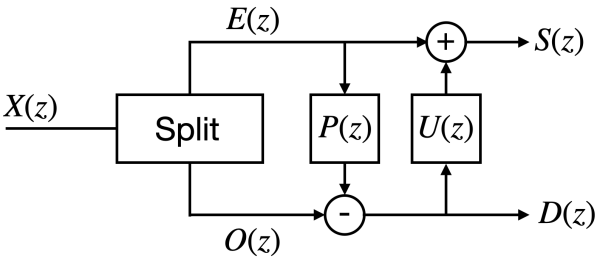


Figure 4a: A schematic plot of lifting scheme in z domain.

(a) Assume $P(z) = \beta(1 + z^{-1})$ and $U(z) = \gamma(1 + z)$, where $\beta, \gamma \in \mathbb{R}$.
The lifting scheme can be written in terms of polyphase matrix as follows:

$$\begin{bmatrix} S(z) \\ D(z) \end{bmatrix} = \begin{bmatrix} H_{0e}(z) & H_{0o}(z) \\ H_{1e}(z) & H_{1o}(z) \end{bmatrix} \begin{bmatrix} E(z) \\ O(z) \end{bmatrix},$$

where $E(z), O(z)$ and $D(z), S(z)$ are the $z-$transform of $x_e[n], x_o[n]$ and $d[n], s[n]$, respectively. Derive the expression for $H_{0e}(z)$, $H_{0o(z)}$, $H_{1e}(z)$ and $H_{1o}(z)$ in terms of $P(z)$ and $U(z)$. [7]

(b) Assume $\beta = \frac{1}{2}$ and $\gamma = 1$, using the given $P(z)$ and $U(z)$, derive the equivalent analysis filters $H_0(z)$ and $H_1(z)$. Note that:

$$H_0(z) = H_{0e}(z^2) + z^{-1}H_{0o}(z^2)$$

and

$$H_1(z) = H_{1e}(z^2) + z^{-1}H_{1o}(z^2).$$

[6]

Question continues on next page

(c) Assume $\beta = \frac{1}{2}$ and $\gamma = 1$ and also assume $x[n] = n$, write the exact expression for $d[n]$ and $s[n]$. [6]

(d) Instead of predicting $x_o[n]$ from two neighboring even samples, we allow a 3-point predictor:

$$d[n] = x_o[n] - (\alpha x_e[n-1] + \beta x_e[n] + \eta x_e[n+1]) .$$

Determine $\alpha, \beta, \eta \in \mathbb{R}$ such that $d[n] = 0$ for all signals of the form

$$x[n] = an^2 + bn + c$$

with $a, b, c \in \mathbb{R}$. [6]